

Unknown Title



Description

Description



Note

Note



Editorial

Editorial



Solutions

Solutions



Submissions

Submissions

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Code

Code



Testcase

Testcase

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Test Result

Test Result



Leet

Leet

×

2. Add Two Numbers

Medium



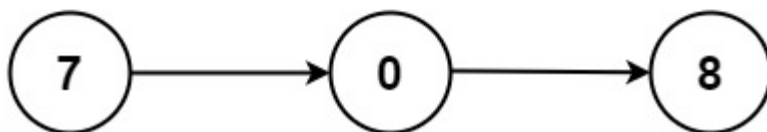
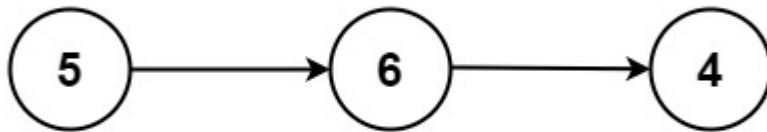
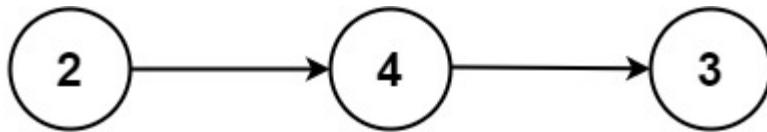
Topics

Companies

You are given two **non-empty** linked lists representing two non-negative integers. The digits are stored in **reverse order**, and each of their nodes contains a single digit. Add the two numbers and return the sum as a linked list.

You may assume the two numbers do not contain any leading zero, except the number 0 itself.

Example 1:



Input: l1 = [2,4,3], l2 = [5,6,4]

Output: [7,0,8]

Explanation: 342 + 465 = 807.

Example 2:

Input: l1 = [0], l2 = [0]

Output: [0]

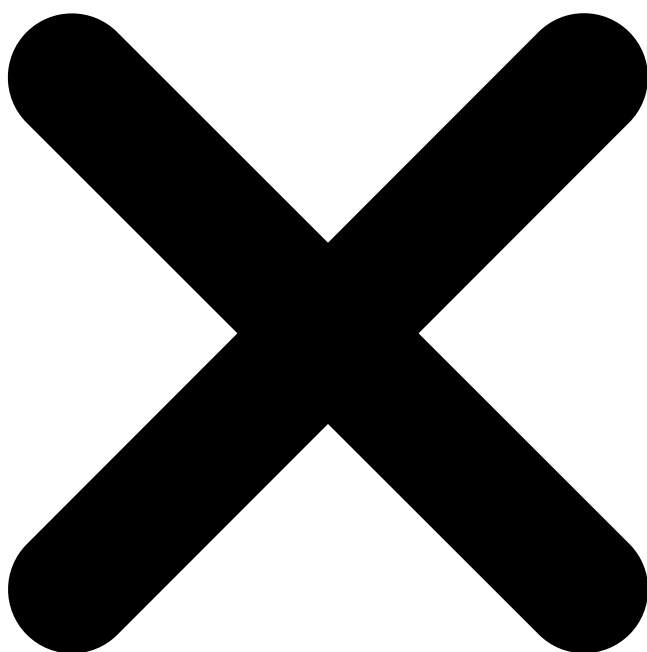
Example 3:

Input: l1 = [9,9,9,9,9,9,9], l2 = [9,9,9,9]

Output: [8,9,9,9,0,0,0,1]

Constraints:

- The number of nodes in each linked list is in the range [1, 100].
- $0 \leq \text{Node.val} \leq 9$
- It is guaranteed that the list represents a number that does not have leading zeros.



Seen this question in a real interview before?

1/6

Yes

No

Accepted

7,137,374/14.7M

Acceptance Rate

48.6%



Companies



Discussion (1.1K)



 Discussion Rules



1. Please don't post **any solutions** in this discussion.
2. The problem discussion is for asking questions about the problem or for sharing tips - anything except for solutions.
3. If you'd like to share your solution for feedback and ideas, please head to the solutions tab and post it there.



dabiggmoe

Nov 21, 2020

I cannot pass this test case

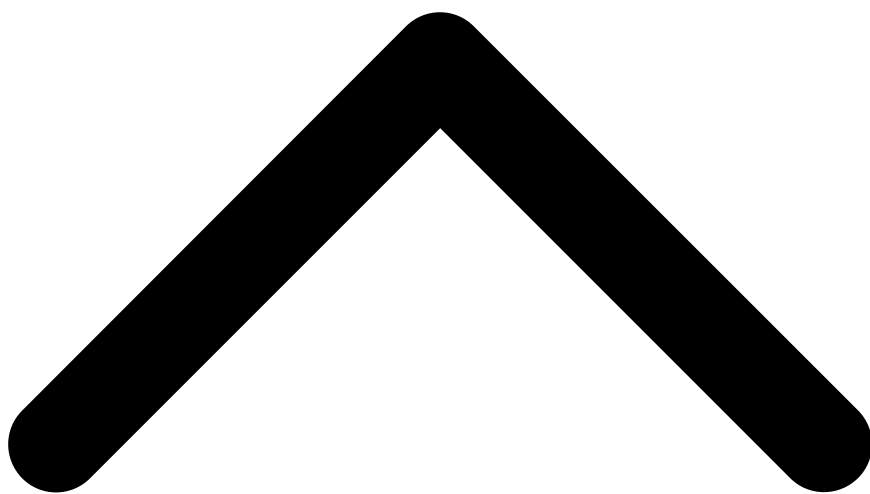
l1 = [2,4,9]

l2 = [5,6,4,9]

my answer is = [8,9,8,5]

But my submission is failing, saying the expected result is [7,0,4,0,1]

Am I missing anything? How can I report this test case?



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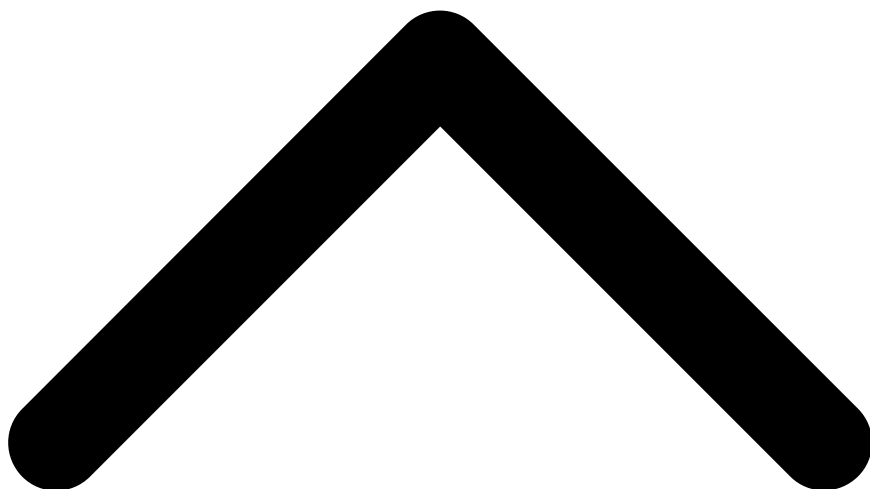


[davidmariluch](#)

Apr 26, 2018

Does anyone else think the ListNode class given is absolute trash? It isn't a Java class in the Java 8 or 9

library. It has no class methods. It's not static. So accessing anything in it is just a crap shoot trial and error?
Who made this trash problem?



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1.1K



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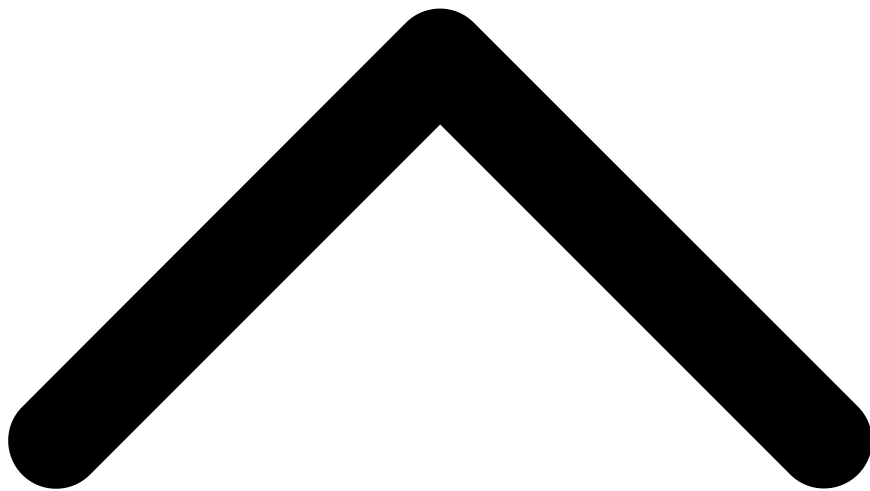
thekip0

Jan 02, 2019

Getting reverse() is not a function in javascript when doing

```
[1,2,3].reverse()
```

but it works in the playground...



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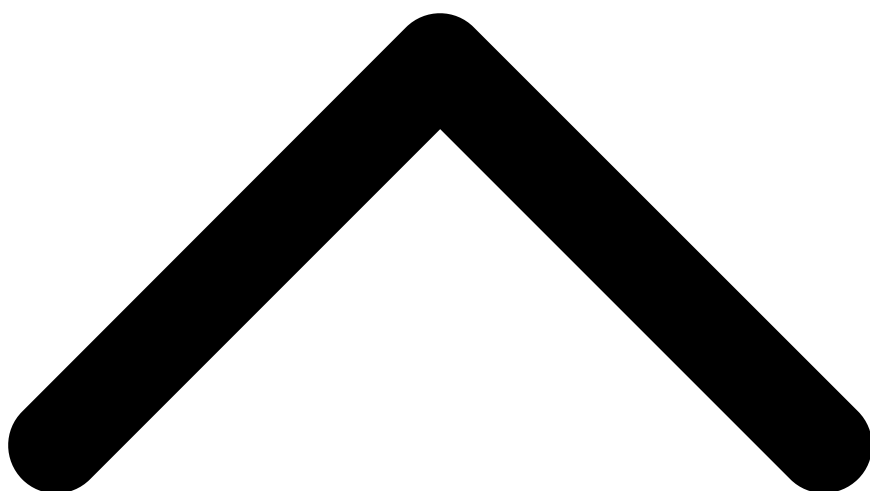
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Anonymous

Jun 02, 2018

The problem description says "You are given two non-empty linked lists representing two non-negative integers". But with the test case that doesn't fit into even Int64 these are clearly NOT integers, even though they are numbers. Please, change the problem description not to mislead anyone and add an appropriate section stating that numbers can contain any amount of digits.



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III

Dec 09, 2018

Why the hell do you have test cases that pass such dumbass strings. How do you expect me to parse a value that long into an Integer? Not even a Long will suffice



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128



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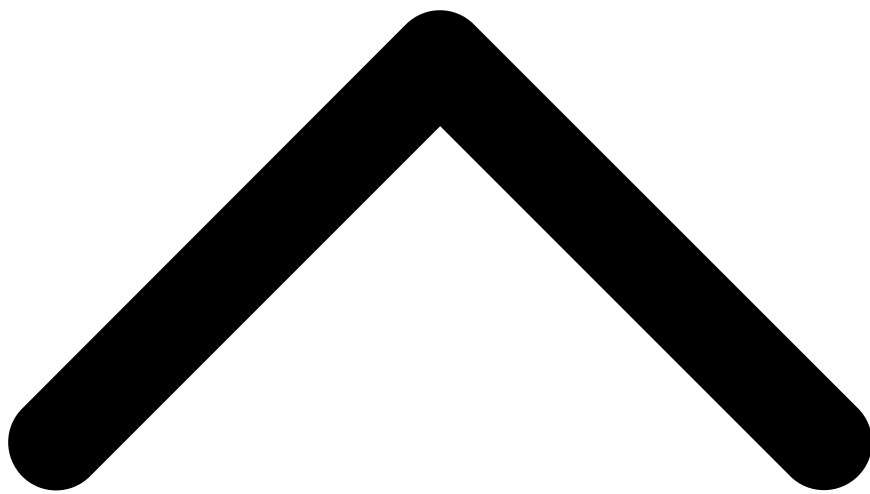
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sonakshi

Jul 20, 2014

Do we need to include the cases when l1 and l2 are of different sizes ?



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97



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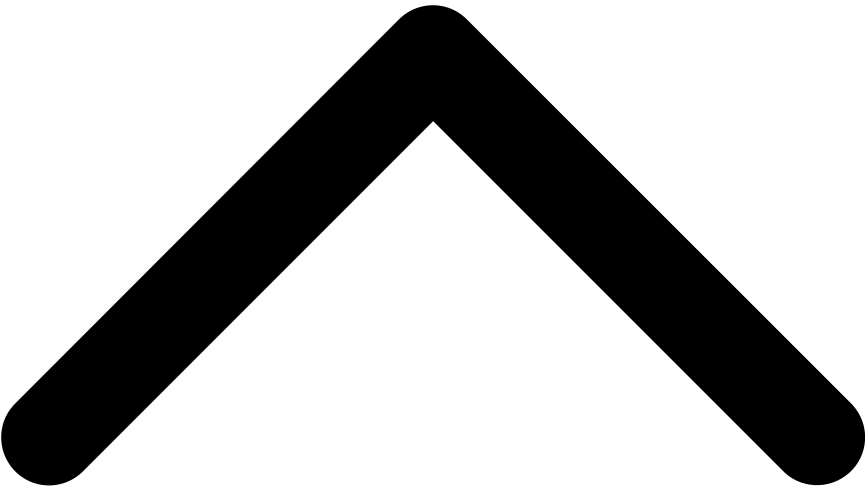


[jerry_lin](#)

Feb 18, 2015

Look at the example input : $(2 \rightarrow 4 \rightarrow 3) + (5 \rightarrow 6 \rightarrow 4)$, why the output is the single link list $7 \rightarrow 0 \rightarrow 8$. How do those

numbers calculated?



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[bwpge](#)

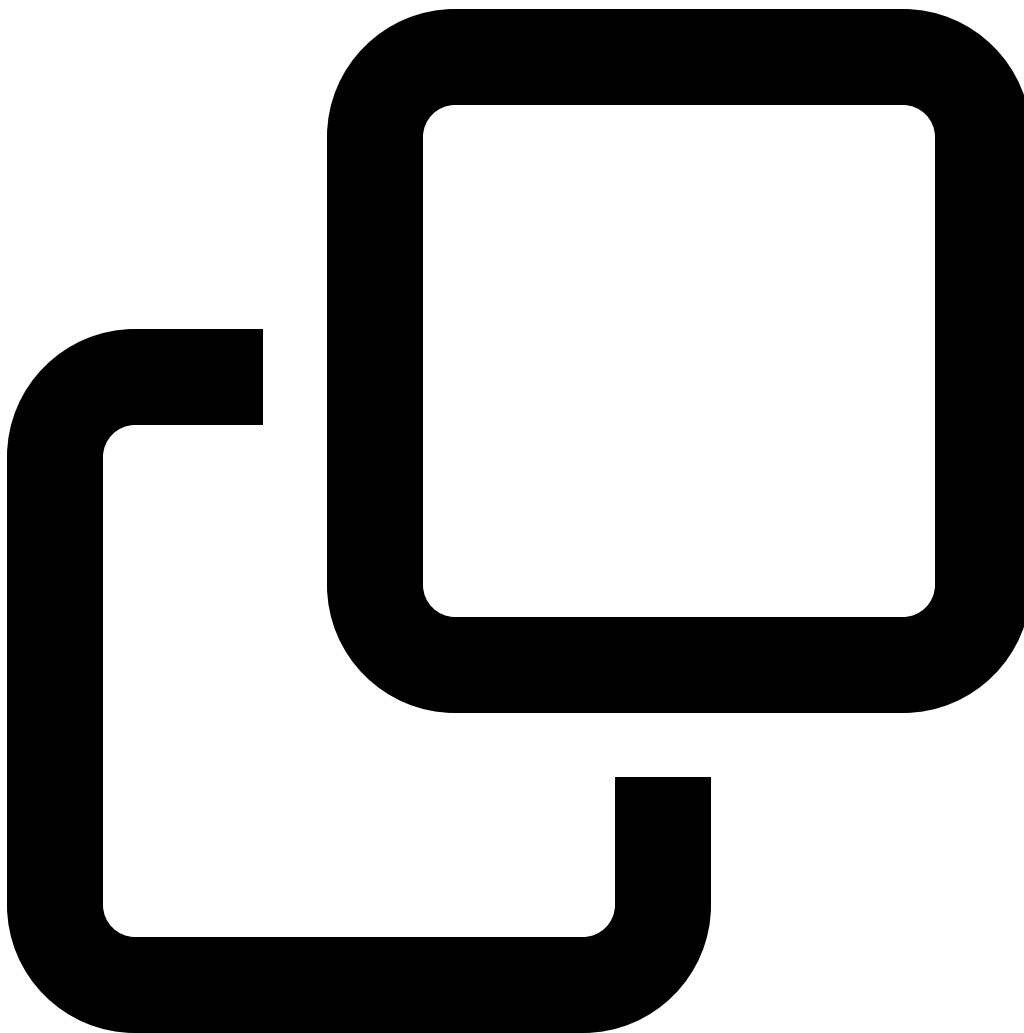


Feb 19, 2023

Seeing some in the comments struggling with parsing the linked list into an integer for your respective language and running into capacity issues (e.g., a number larger than 32-bit or 64-bit integer can hold). **This is the wrong approach.**

Think of how you learned to sum large numbers on scratch paper. Consider how you look at each column and handle carries for the next column:

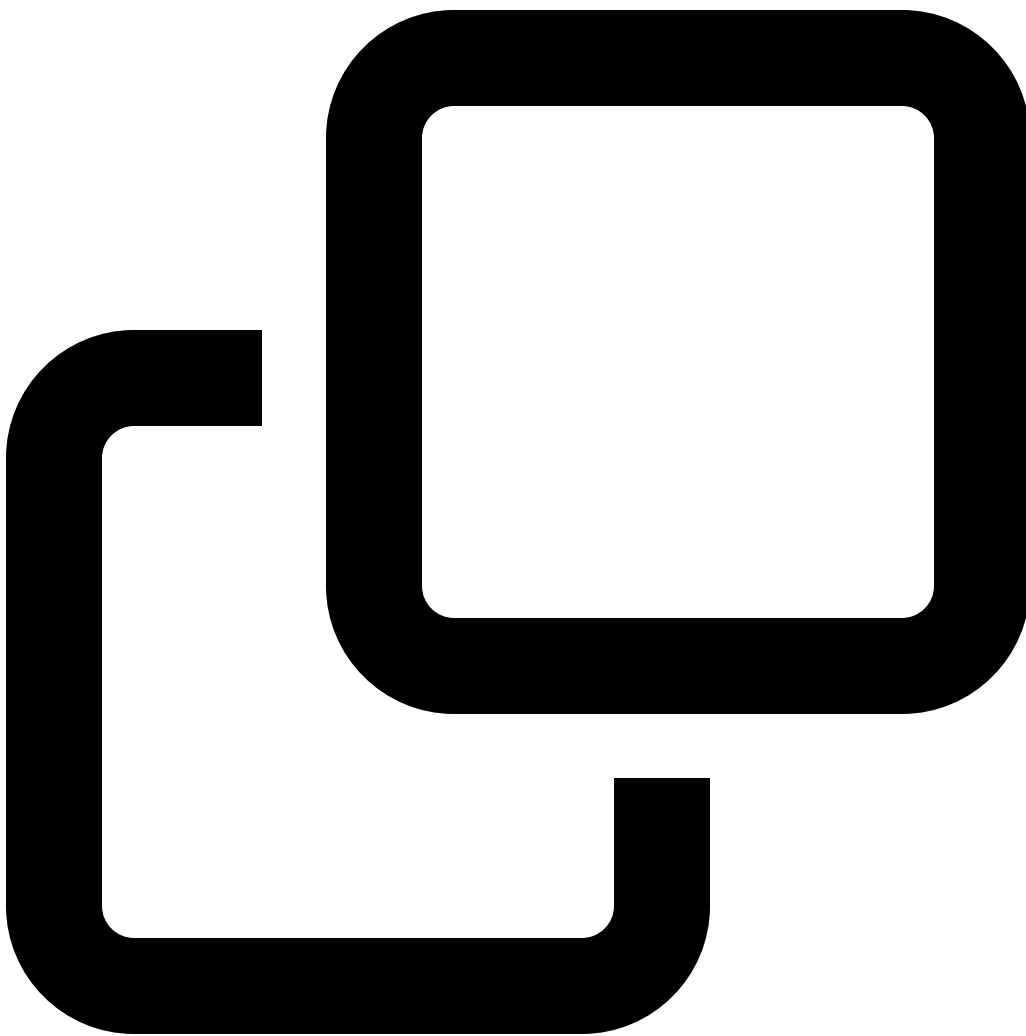
```
  1 1 1
 1 2 3 4 5
+   7 8 9
-----
 1 3 1 3 4
```



Now if you treat these as a linked list of numbers, starting with the least significant digit:

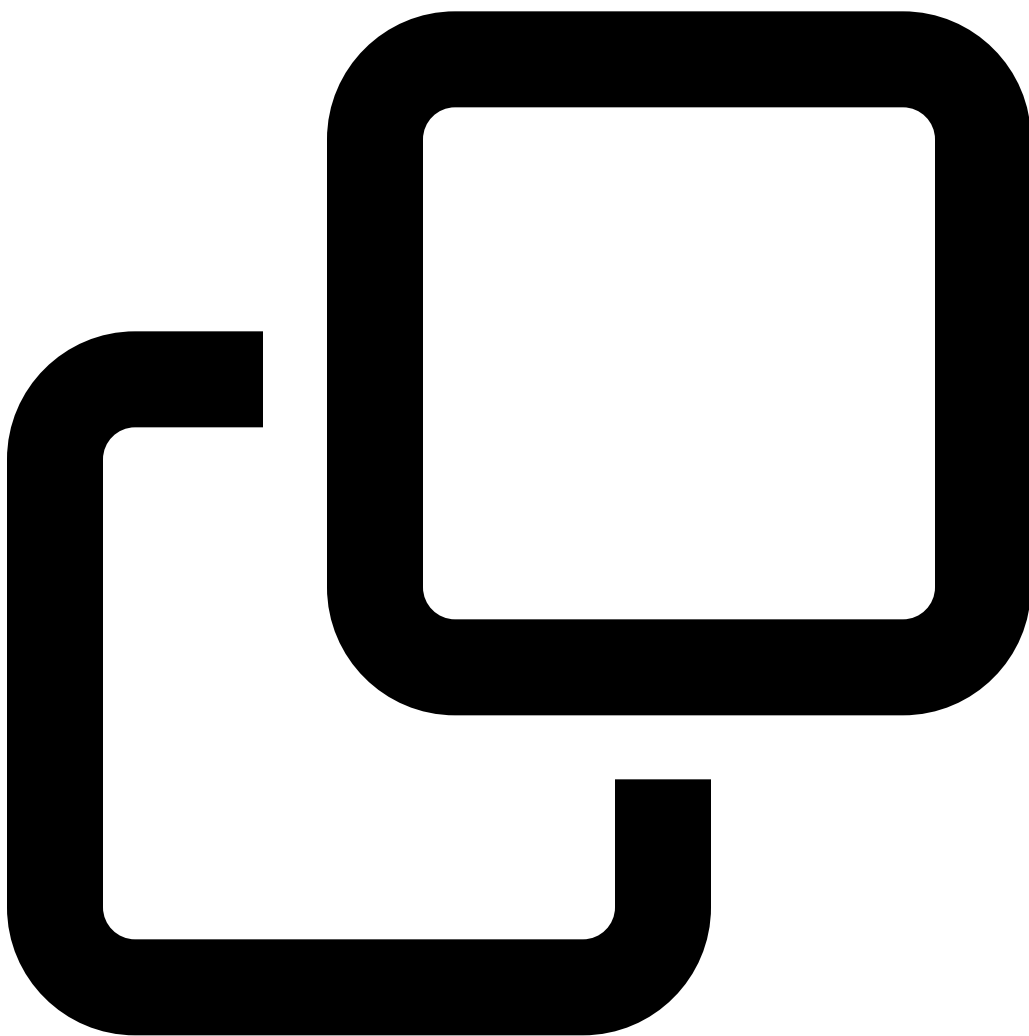
```
      root node ---v
11: 1 <- 2 <- 3 <- 4 <- 5
```

```
12:          7 <- 8 <- 9
      root node ---^
```



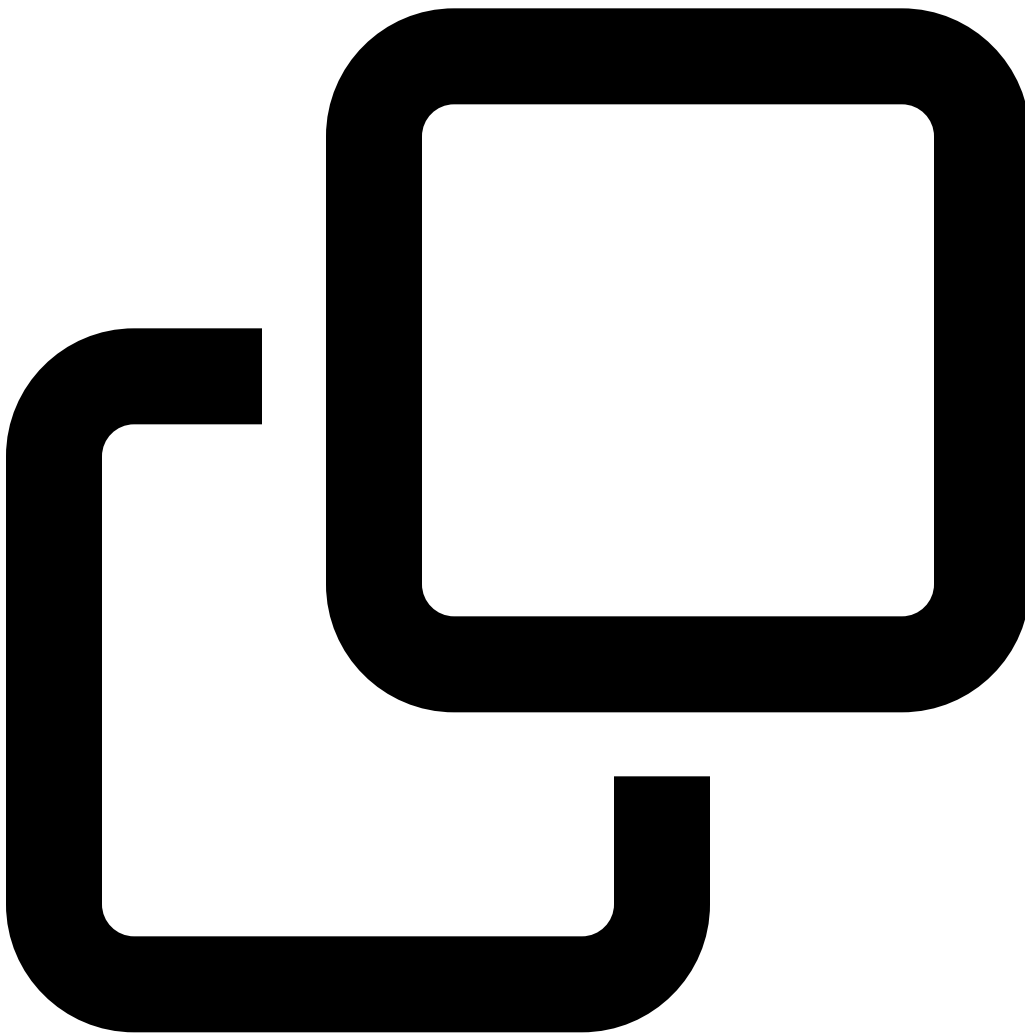
Flipping these lists to order left to right like in the problem statement:

```
v--- root node
11: 5 -> 4 -> 3 -> 2 -> 1
12: 9 -> 8 -> 7
    ^--- root node
```



You can step through each node in order, and simply treat a missing node as 0 (like we do in scratch paper above):

```
11: 5 -> 4 -> 3 -> 2 -> 1
12: 9 -> 8 -> 7   (0) (0)
-----
a:  4 -> 3 -> 1 -> 3 -> 1
```



Be sure to keep track of carry over into next step (T below means true, F means false):

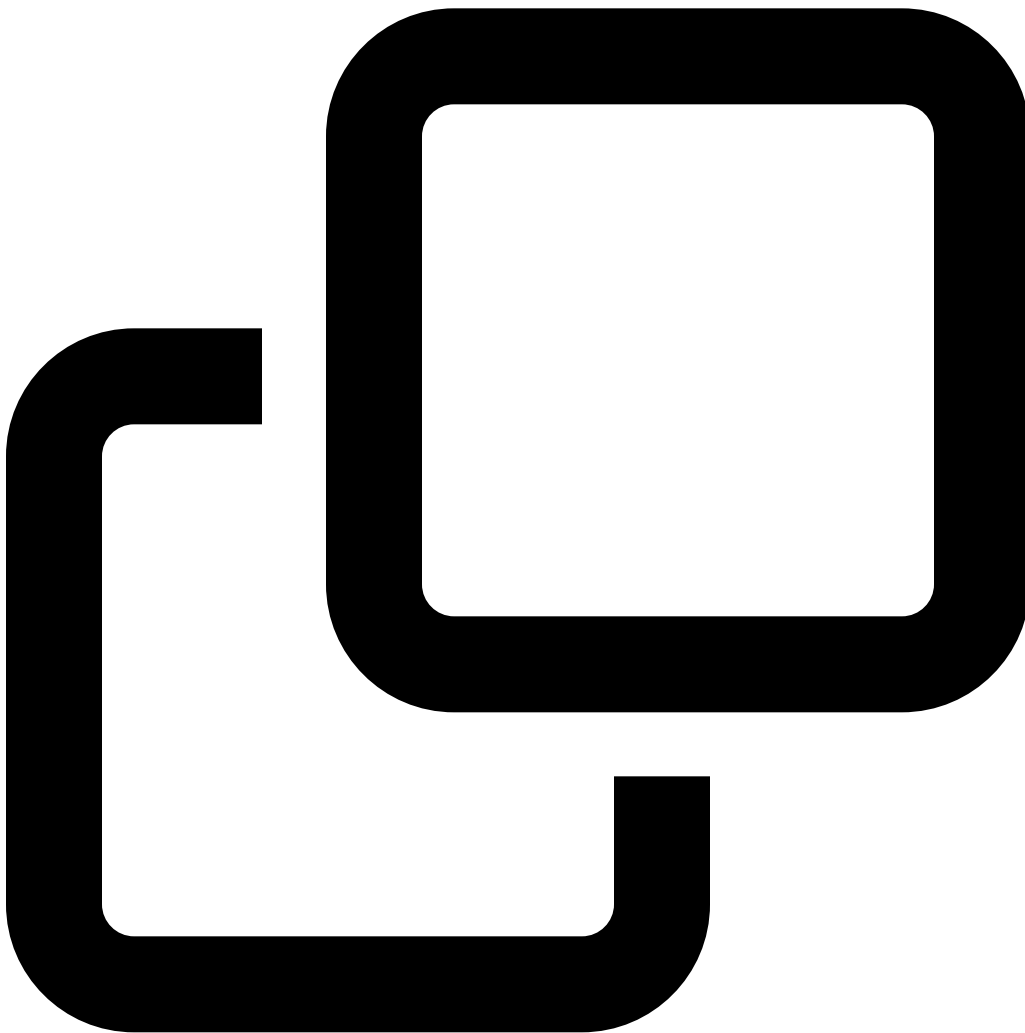
has carry:

F T T T F

l1: 5 -> 4 -> 3 -> 2 -> 1

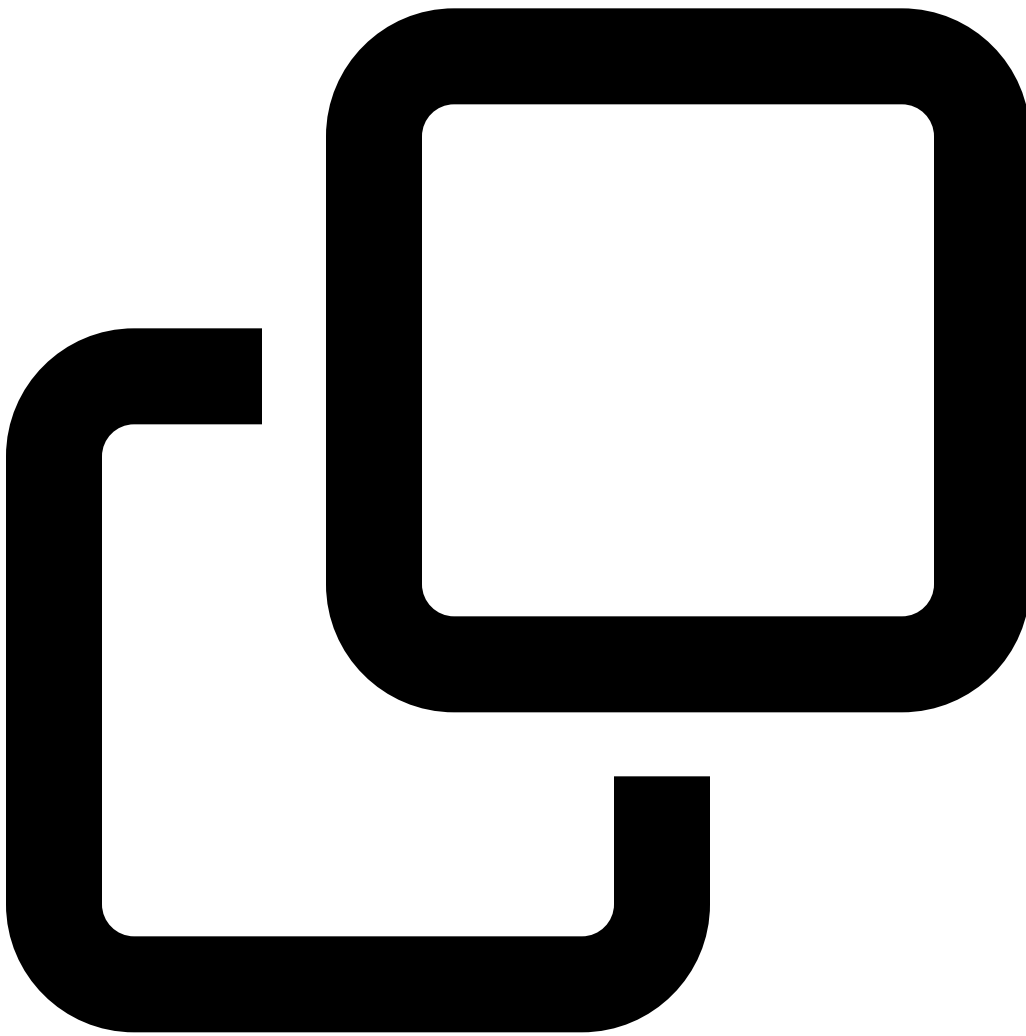
l2: 9 -> 8 -> 7 (0) (0)

a: 4 -> 3 -> 1 -> 3 -> 1



As a final note, consider what happens if there is a carry when there are no more nodes in the lists, such as with:

```
has carry:
  F   T   T   T   T   T
-----
11: 9 -> 9 -> 9 -> 9 -> 9   (?)
12: 9 -> 9 -> 9   (0)  (0)  (?)
-----
a: 8 -> 9 -> 9 -> 0 -> 0 -> 1
```



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53



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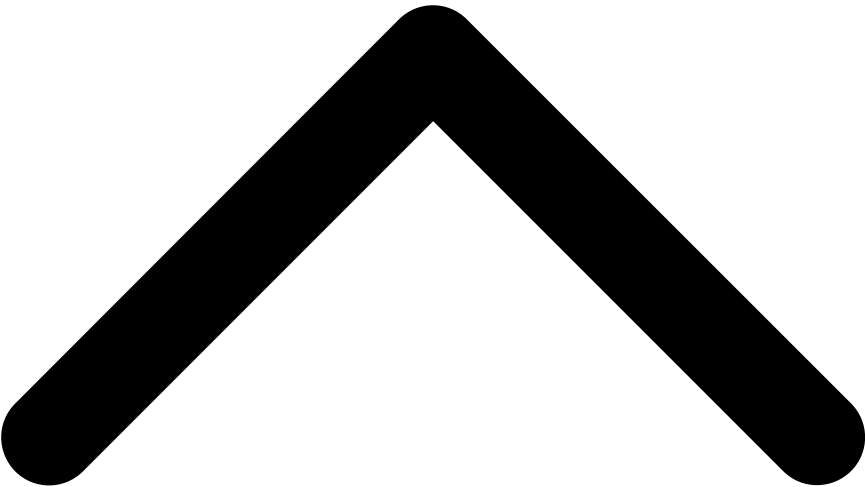
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Yevgen Safronov

Jun 14, 2016

From the task definition, it is clear that we should return a linked list.



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641 Online

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

/**

* Definition for singly-linked list.

* public class ListNode {

* int val;

* ListNode next;

* ListNode() {}

* ListNode(int val) { this.val = val; }

* ListNode(int val, ListNode next) { this.val = val; this.next = next; }

* }

```
*/  
  
class Solution {  
  
    public ListNode addTwoNumbers(ListNode l1, ListNode l2) {  
  
  
    }  
}
```



Saved

Ln 1, Col 1

You must run your code first



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